

Multiple Choice

1. **Answer:** A

Options: A) Temperature, humidity, and human activities; B) Only soil structure and chemistry; C) Only sunlight and humidity; D) Sunlight, soil pH, and predator-prey relationships; E) Only temperature and precipitation

Note: Correct option: A - Temperature, humidity, and human activities

2. **Answer:** B

Options: A) the enzyme is inactive at all pH values; B) no ionizing groups on the enzyme or substrate participate in the reaction; C) only ionizing groups on the substrate participate in the reaction; D) ionizing groups on both the enzyme and substrate participate in the reaction; E) only non-ionizing groups on the substrate participate in the reaction

Note: Correct option: B - no ionizing groups on the enzyme or substrate participate in the reaction

3. **Answer:** B

Options: A) Cyanobacteria reduce sulfur compounds.; B) Cyanobacteria have no nuclei.; C) Green algae cannot reduce nitrogen compounds.; D) Green algae cannot photosynthesize.; E) Green algae produce cell membranes.

Note: Correct option: B - Cyanobacteria have no nuclei.

4. **Answer:** B

Options: A) Simple diffusion can only transport minerals when they are bound to oxygen molecules; B) Simple diffusion cannot occur against a concentration gradient; C) Simple diffusion requires specific carrier proteins for each mineral; D) Simple diffusion only works for organic compounds, not inorganic minerals; E) Simple diffusion can transport minerals in any concentration

Note: Correct option: B - Simple diffusion cannot occur against a concentration gradient

5. **Answer:** D

Options: A) Lake B is alkaline.; B) Lake B is 100 times more alkaline than Lake A.; C) The hydrogen ion concentration of Lake B is 10^{-4} M.; D) The pH of both lakes will decrease in response to acid rain.

Note: Correct option: D - The pH of both lakes will decrease in response to acid rain.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: People presenting with type 2 diabetes with lower initial glycemia who may be earlier in the course of their disease had fewer adverse clinical outcomes despite similar glycemic progression. Since most such people are asymptomatic at diagnosis, active case detection programs would be required to identify them.

7. **Answer:** False

Options: A) True; B) False

Note: According to the results of our study and the data from the international literature, the entity globulomaxillary cyst is no longer justified.

8. **Answer:** False

Options: A) True; B) False

Note: Late-night salivary cortisol has a poor specificity for cortisol excess in obese patients with T 2 DM with 24 hr UFC showing significantly better specificity in our population.

9. **Answer:** False

Options: A) True; B) False

Note: The study suggests that there is no difference in clinical toxicity profiles or survival outcomes with either definitive chemoradiotherapy or chemoradiation followed by surgery in management of locally advanced esophageal cancer.

10. **Answer:** False

Options: A) True; B) False

Note: Our data showed no difference in 25(OH) D levels between normal children and those with CD when adjusted for body mass index.

Long Form Response

11. **Answer:** 1/16. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Inbreeding and loss of genetic variation threaten a population.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key